Missing Value Check - Raw Data

Variable	N Miss	N
litter	0	79
trt	0	79
wgtgain	4	75

Range Check - Weight Gain Variable (Raw)

Analysis Variable : wgtgain				
Minimum	Maximum	Mean	Std Dev	N
-3.2000000	99.5000000	21.6069333	9.7298128	75

Frequency Check - Treatment Groups and Litter IDs (Raw)

trt	Frequency	Percent
1	9	11.39
2	9	11.39
3	10	12.66
4	8	10.13
5	8	10.13
6	8	10.13
7	9	11.39
8	9	11.39
9	8	10.13
10	1	1.27

litter	Frequency	Percent
1	9	11.39
2	10	12.66
3	10	12.66
4	10	12.66
5	11	13.92
6	10	12.66
7	9	11.39
8	9	11.39
9	1	1.27

Duplicate Records Found

Obs	litter	trt	wgtgain
1	2	1	19.86
2	4	7	99.5
3	5	3	23.96
4	5	8	.
5	6	3	-3.2

Verification - Cleaned Dataset

Variable	N Miss	N	Minimum	Maximum	Mean	Std Dev
litter	0	69	1.0000000	8.0000000	4.5072464	2.3491124
trt	0	69	1.0000000	9.0000000	4.9855072	2.6205789
wgtgain	0	69	16.4400000	26.3800000	20.9268116	1.9526727

Frequency Check - Treatment Groups and Litter IDs (Cleaned)

trt	Frequency	Percent
1	8	11.59
2	8	11.59
3	7	10.14
4	8	11.59
5	7	10.14
6	8	11.59
7	8	11.59
8	7	10.14
9	8	11.59

litter	Frequency	Percent
1	9	13.04
2	9	13.04
3	8	11.59
4	9	13.04
5	7	10.14
6	9	13.04
7	9	13.04
8	9	13.04

Frequency Check - Treatment Groups and Litter IDs (Cleaned)

Analysis Variable : wgtgain						
trt	N Obs	Mean	Std Dev	Minimum	Maximum	N
1	8	22.4175000	2.4471076	18.7500000	26.3800000	8
2	8	21.5425000	1.7331289	19.1300000	23.9500000	8
3	7	20.3428571	2.3022287	17.3000000	23.9600000	7
4	8	19.9637500	0.7482539	19.0200000	21.1600000	8
5	7	20.7771429	2.2874419	16.4400000	23.7100000	7
6	8	21.9462500	1.9658218	18.6800000	25.0400000	8
7	8	19.9650000	1.5524819	17.3800000	21.9800000	8
8	7	19.8900000	1.3195201	18.5200000	22.5400000	7
9	8	21.2750000	1.6846958	18.1400000	23.1400000	8

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

Treatment	N	Mean	Std Dev	Minimum	Maximum
1	8	22.418	2.447	18.750	26.380
2	8	21.542	1.733	19.130	23.950
3	7	20.343	2.302	17.300	23.960
4	8	19.964	0.748	19.020	21.160
5	7	20.777	2.287	16.440	23.710
6	8	21.946	1.966	18.680	25.040
7	8	19.965	1.552	17.380	21.980
8	7	19.890	1.320	18.520	22.540
9	8	21.275	1.685	18.140	23.140

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

Model Information	
Data Set	WORK.WEIGHTDATA
Dependent Variable	wgtgain
Covariance Structure	Variance Components
Subject Effect	litter
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Kenward-Roger
Degrees of Freedom Method	Kenward-Roger

Class Level Information		
Class	**Levels**	**Values**
trt	9	1 2 3 4 5 6 7 8 9
litter	8	1 2 3 4 5 6 7 8

Dimensions	
Covariance Parameters	2
Columns in X	10
Columns in Z per Subject	1
Subjects	8
Max Obs per Subject	9

Number of Observations	
Number of Observations Read	69
Number of Observations Used	69
Number of Observations Not Used	0

Iteration History			
Iteration	**Evaluations**	**-2 Res Log Like**	**Criterion**
0	1	262.10063124	
1	2	254.09555581	0.00000028
2	1	254.09553552	0.00000000

Convergence criteria met.

Covariance Parameter Estimates		
Cov Parm	**Subject**	**Estimate**
Intercept	litter	0.8822
Residual		2.5368

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

Fit Statistics	
-2 Res Log Likelihood	254.1
AIC (Smaller is Better)	258.1
AICC (Smaller is Better)	258.3
BIC (Smaller is Better)	258.3

Solution for Fixed Effects

Effect	trt	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		21.2750	0.6537	39.6	32.54	<.0001
trt	1	1.1425	0.7964	52.9	1.43	0.1573
trt	2	0.2675	0.7964	52.9	0.34	0.7383
trt	3	-0.9310	0.8274	53.1	-1.13	0.2656
trt	4	-1.3113	0.7964	52.9	-1.65	0.1056
trt	5	-0.3414	0.8278	53.2	-0.41	0.6817
trt	6	0.6712	0.7964	52.9	0.84	0.4031
trt	7	-1.3100	0.7964	52.9	-1.64	0.1059
trt	8	-1.2285	0.8278	53.2	-1.48	0.1437
trt	9	0	.	.	.	.

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
trt	8	53.1	2.59	0.0182

Least Squares Means

Effect	trt	Estimate	Standard Error	DF	t Value	Pr > |t|
trt	1	22.4175	0.6537	39.6	34.29	<.0001
trt	2	21.5425	0.6537	39.6	32.95	<.0001
trt	3	20.3440	0.6912	43.6	29.43	<.0001
trt	4	19.9637	0.6537	39.6	30.54	<.0001
trt	5	20.9336	0.6917	43.5	30.27	<.0001
trt	6	21.9463	0.6537	39.6	33.57	<.0001
trt	7	19.9650	0.6537	39.6	30.54	<.0001
trt	8	20.0465	0.6917	43.5	28.98	<.0001
trt	9	21.2750	0.6537	39.6	32.54	<.0001

Table 1. Descriptive Statistics of Weight Gain by Treatment Group

Differences of Least Squares Means									
Effect	**trt**	**_trt**	**Estimate**	**Standard Error**	**DF**	**t Value**	**Pr > |t|**	**Adjustment**	**Adj P**
trt	1	2	0.8750	0.7964	52.9	1.10	0.2769	Tukey-Kramer	0.9719
trt	1	3	2.0735	0.8274	53.1	2.51	0.0153	Tukey-Kramer	0.2526
trt	1	4	2.4537	0.7964	52.9	3.08	0.0033	Tukey-Kramer	0.0731
trt	1	5	1.4839	0.8278	53.2	1.79	0.0787	Tukey-Kramer	0.6866
trt	1	6	0.4713	0.7964	52.9	0.59	0.5565	Tukey-Kramer	0.9996
trt	1	7	2.4525	0.7964	52.9	3.08	0.0033	Tukey-Kramer	0.0733
trt	1	8	2.3710	0.8278	53.2	2.86	0.0060	Tukey-Kramer	0.1213
trt	1	9	1.1425	0.7964	52.9	1.43	0.1573	Tukey-Kramer	0.8791
trt	2	3	1.1985	0.8274	53.1	1.45	0.1534	Tukey-Kramer	0.8734
trt	2	4	1.5788	0.7964	52.9	1.98	0.0526	Tukey-Kramer	0.5619
trt	2	5	0.6089	0.8278	53.2	0.74	0.4652	Tukey-Kramer	0.9980
trt	2	6	-0.4037	0.7964	52.9	-0.51	0.6143	Tukey-Kramer	0.9999
trt	2	7	1.5775	0.7964	52.9	1.98	0.0528	Tukey-Kramer	0.5629
trt	2	8	1.4960	0.8278	53.2	1.81	0.0764	Tukey-Kramer	0.6772
trt	2	9	0.2675	0.7964	52.9	0.34	0.7383	Tukey-Kramer	1.0000
trt	3	4	0.3803	0.8274	53.1	0.46	0.6477	Tukey-Kramer	0.9999
trt	3	5	-0.5896	0.8587	53.5	-0.69	0.4953	Tukey-Kramer	0.9988
trt	3	6	-1.6022	0.8274	53.1	-1.94	0.0581	Tukey-Kramer	0.5926
trt	3	7	0.3790	0.8274	53.1	0.46	0.6488	Tukey-Kramer	0.9999
trt	3	8	0.2976	0.8587	53.5	0.35	0.7303	Tukey-Kramer	1.0000
trt	3	9	-0.9310	0.8274	53.1	-1.13	0.2656	Tukey-Kramer	0.9677
trt	4	5	-0.9699	0.8278	53.2	-1.17	0.2466	Tukey-Kramer	0.9590
trt	4	6	-1.9825	0.7964	52.9	-2.49	0.0160	Tukey-Kramer	0.2604
trt	4	7	-0.00125	0.7964	52.9	-0.00	0.9988	Tukey-Kramer	1.0000
trt	4	8	-0.08271	0.8278	53.2	-0.10	0.9208	Tukey-Kramer	1.0000
trt	4	9	-1.3113	0.7964	52.9	-1.65	0.1056	Tukey-Kramer	0.7748
trt	5	6	-1.0126	0.8278	53.2	-1.22	0.2266	Tukey-Kramer	0.9476
trt	5	7	0.9686	0.8278	53.2	1.17	0.2472	Tukey-Kramer	0.9593
trt	5	8	0.8871	0.8513	52.9	1.04	0.3021	Tukey-Kramer	0.9797
trt	5	9	-0.3414	0.8278	53.2	-0.41	0.6817	Tukey-Kramer	1.0000
trt	6	7	1.9812	0.7964	52.9	2.49	0.0160	Tukey-Kramer	0.2612
trt	6	8	1.8998	0.8278	53.2	2.30	0.0257	Tukey-Kramer	0.3635
trt	6	9	0.6712	0.7964	52.9	0.84	0.4031	Tukey-Kramer	0.9949
trt	7	8	-0.08146	0.8278	53.2	-0.10	0.9220	Tukey-Kramer	1.0000
trt	7	9	-1.3100	0.7964	52.9	-1.64	0.1059	Tukey-Kramer	0.7757
trt	8	9	-1.2285	0.8278	53.2	-1.48	0.1437	Tukey-Kramer	0.8578

Table 2. Covariance Parameter Estimates (Random Effect of Litter)

Covariance Parameter	Subject	Estimate
Intercept	litter	0.8822
Residual		2.5368

Table 3. Fixed Effects Estimates - PROC MIXED (REML)

Effect	Treatment	Estimate	Std Error	DF	t Value	p Value
Intercept	_	21.2750	0.6537	39.6	32.54	0.0000
trt	1	1.1425	0.7964	52.9	1.43	0.1573
trt	2	0.2675	0.7964	52.9	0.34	0.7383
trt	3	-0.9310	0.8274	53.1	-1.13	0.2656
trt	4	-1.3113	0.7964	52.9	-1.65	0.1056
trt	5	-0.3414	0.8278	53.2	-0.41	0.6817
trt	6	0.6712	0.7964	52.9	0.84	0.4031
trt	7	-1.3100	0.7964	52.9	-1.64	0.1059
trt	8	-1.2285	0.8278	53.2	-1.48	0.1437
trt	9	0.0000	.	.	.	.

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Effect	Num DF	Den DF	F Value	p Value
trt	8	53.1	2.589	0.0182

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Class Level Information		
Class	**Levels**	**Values**
trt	9	1 2 3 4 5 6 7 8 9
litter	8	1 2 3 4 5 6 7 8

Number of Observations Read	69
Number of Observations Used	69

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Dependent Variable: wgtgain

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	15	124.9518071	8.3301205	3.29	0.0007
Error	53	134.3274915	2.5344810		
Corrected Total	68	259.2792986			

R-Square	Coeff Var	Root MSE	wgtgain Mean
0.481920	7.607491	1.592005	20.92681

Source	DF	Type I SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.71	0.0140
litter	7	69.96886926	9.99555275	3.94	0.0016

Source	DF	Type III SS	Mean Square	F Value	Pr > F
trt	8	51.75790299	6.46973787	2.55	0.0197
litter	7	69.96886926	9.99555275	3.94	0.0016

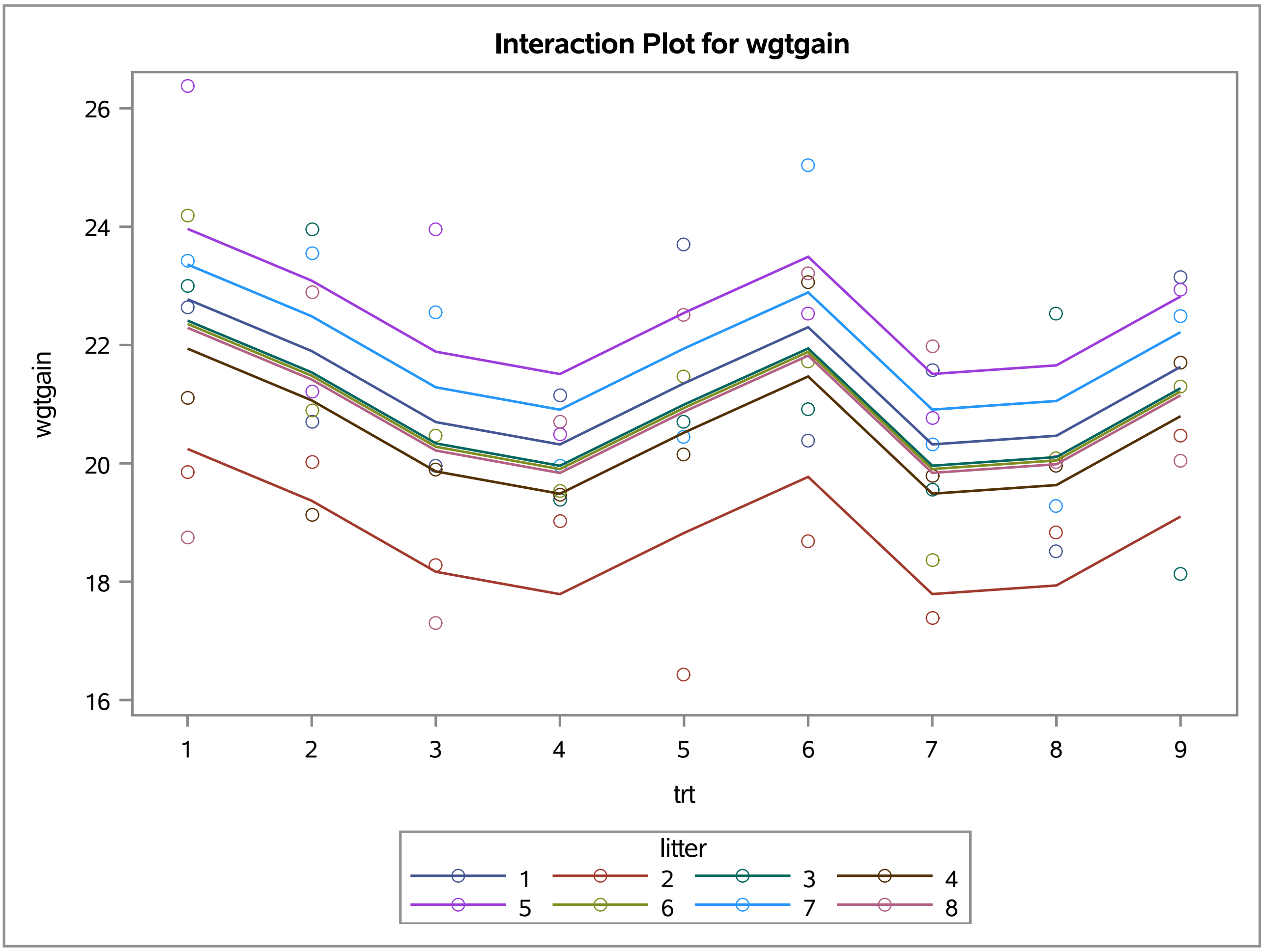


Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Variable: resid

Moments			
N	69	Sum Weights	69
Mean	0	Sum Observations	0
Std Deviation	1.40549076	Variance	1.97540429
Skewness	-0.2389607	Kurtosis	-0.289214
Uncorrected SS	134.327491	Corrected SS	134.327491
Coeff Variation	.	Std Error Mean	0.1692012

Basic Statistical Measures			
Location		Variability	
Mean	0.000000	Std Deviation	1.40549
Median	0.034878	Variance	1.97540
Mode	.	Range	5.97591
		Interquartile Range	2.01125

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	1.5	Pr >= |M|	0.8099
Signed Rank	S	22.5	Pr >= |S|	0.8941

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.980992	Pr < W	0.3761
Kolmogorov-Smirnov	D	0.05578	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.030027	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.248981	Pr > A-Sq	>0.2500

Quantiles (Definition 5)	
Level	Quantile
100% Max	2.4340638
99%	2.4340638
95%	2.3566243
90%	2.0724289
75% Q3	1.0642615
50% Median	0.0348783
25% Q1	-0.9469885
10%	-1.9217107
5%	-2.3833757

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

Variable: resid

Quantiles (Definition 5)	
Level	**Quantile**
1%	-3.5418496
0% Min	-3.5418496

Extreme Observations			
Lowest		**Highest**	
Value	**Obs**	**Value**	**Obs**
-3.54185	61	2.15051	57
-3.13016	26	2.35662	5
-2.91652	63	2.41234	20
-2.38338	14	2.41710	36
-1.94623	8	2.43406	25

Table 4. Type 3 Test of Fixed Effects - Overall Treatment Effect

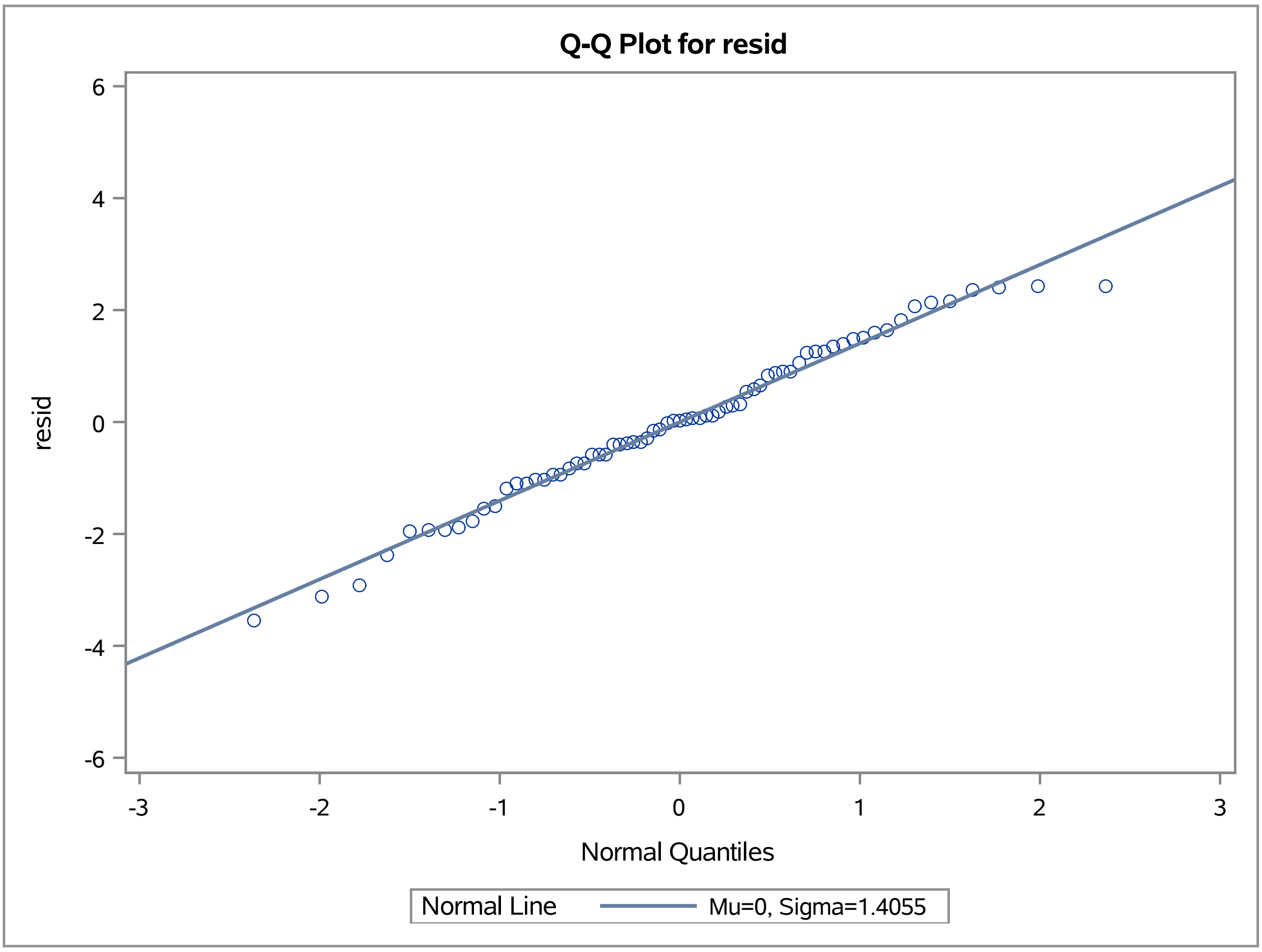


Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

Test	Statistic	p Value
Shapiro-Wilk	0.980992	0.3761
Kolmogorov-Smirnov	0.05578	0.1500
Cramer-von Mises	0.030027	0.2500
Anderson-Darling	0.248981	0.2500

Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

Class Level Information		
Class	**Levels**	**Values**
trt	9	1 2 3 4 5 6 7 8 9

Number of Observations Read	69
Number of Observations Used	69

Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

Dependent Variable: wgtgain

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	54.9829378	6.8728672	2.02	0.0593
Error	60	204.2963607	3.4049393		
Corrected Total	68	259.2792986			

R-Square	Coeff Var	Root MSE	wgtgain Mean
0.212061	8.817625	1.845248	20.92681

Source	DF	Type I SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.02	0.0593

Source	DF	Type III SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.02	0.0593

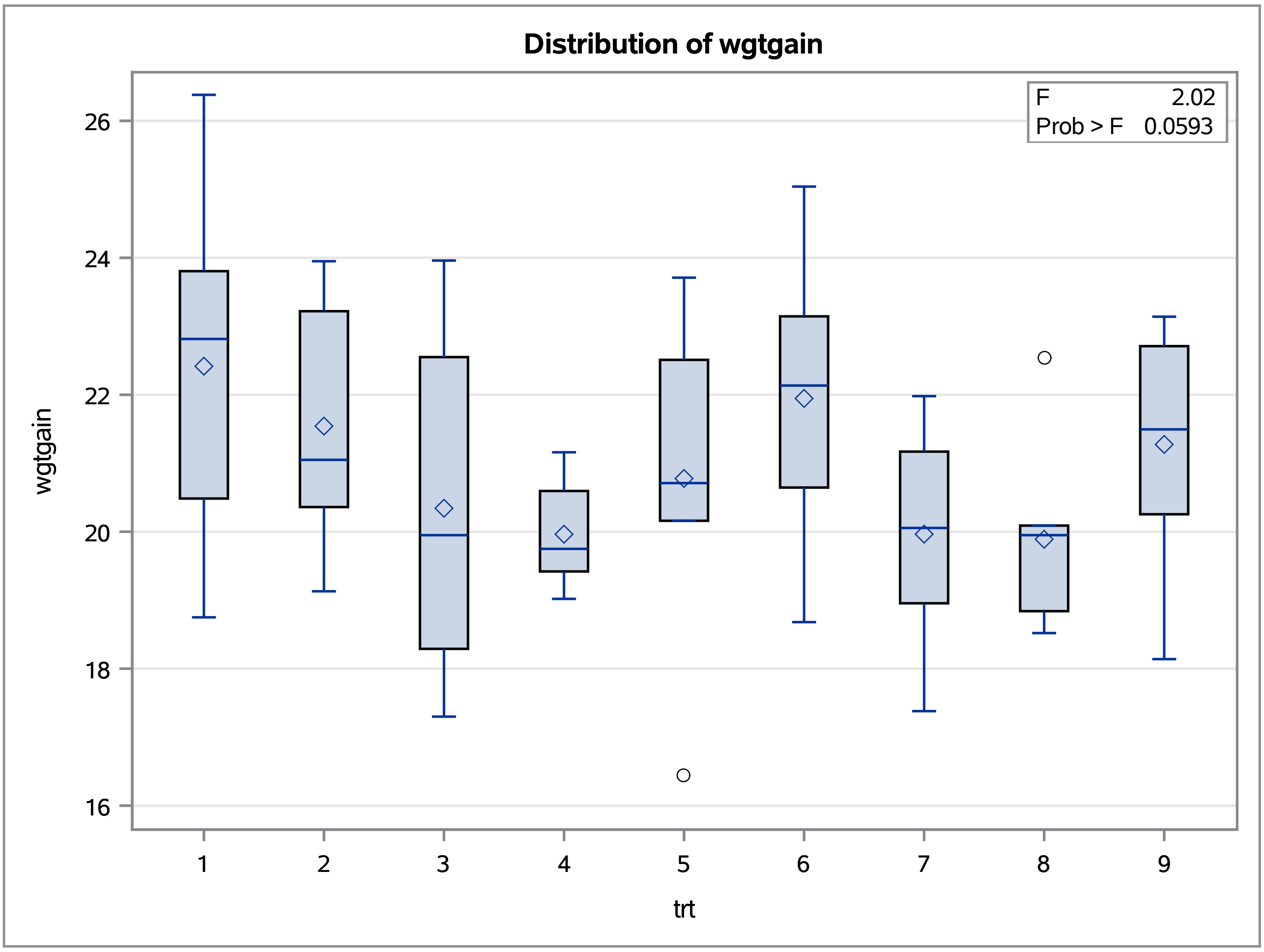
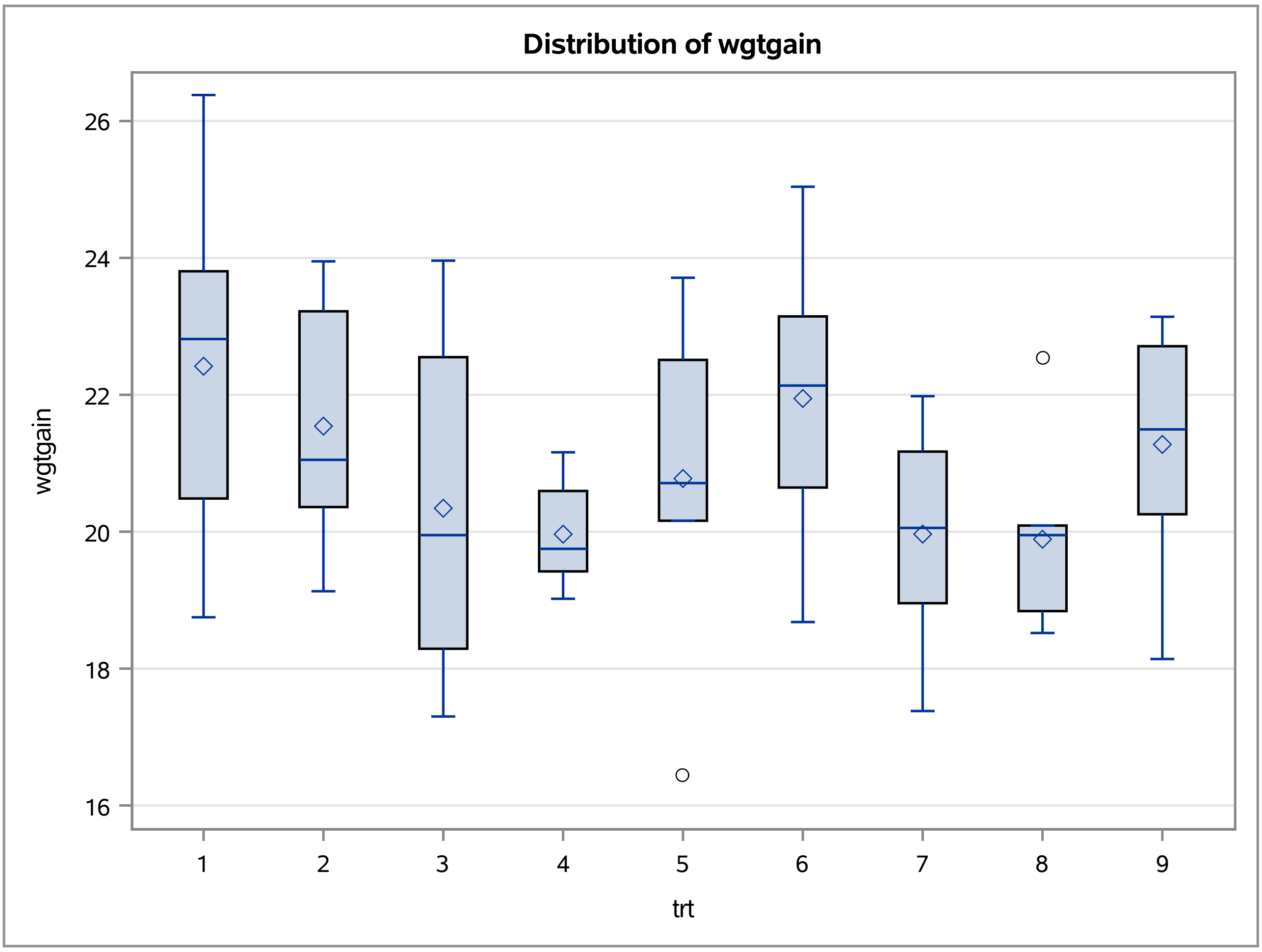


Table 5. Tests for Normality of Residuals (Shapiro-Wilk)

Levene's Test for Homogeneity of wgtgain Variance ANOVA of Squared Deviations from Group Means					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
trt	8	149.2	18.6501	1.09	0.3845
Error	60	1029.2	17.1540		

Table 5. Tests for Normality of Residuals (Shapiro-Wilk)



Level of trt	N	wgtgain Mean	wgtgain Std Dev
1	8	22.4175000	2.44710762
2	8	21.5425000	1.73312888
3	7	20.3428571	2.30222873
4	8	19.9637500	0.74825392
5	7	20.7771429	2.28744191
6	8	21.9462500	1.96582180
7	8	19.9650000	1.55248188
8	7	19.8900000	1.31952011
9	8	21.2750000	1.68469582

Table 6. Levene's Test for Homogeneity of Variance

Source	DF	Sum of Squares	Mean Square	F Value	p Value
trt	8	149.2	18.6501	1.087	0.3845
Error	60	1029.2	17.1540	_	_

Table 6. Levene's Test for Homogeneity of Variance

Class Level Information		
Class	**Levels**	**Values**
trt	9	1 2 3 4 5 6 7 8 9
litter	8	1 2 3 4 5 6 7 8

Number of Observations Read	69
Number of Observations Used	69

Table 6. Levene's Test for Homogeneity of Variance

Dependent Variable: wgtgain

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	15	124.9518071	8.3301205	3.29	0.0007
Error	53	134.3274915	2.5344810		
Corrected Total	68	259.2792986			

R-Square	Coeff Var	Root MSE	wgtgain Mean
0.481920	7.607491	1.592005	20.92681

Source	DF	Type I SS	Mean Square	F Value	Pr > F
trt	8	54.98293784	6.87286723	2.71	0.0140
litter	7	69.96886926	9.99555275	3.94	0.0016

Source	DF	Type III SS	Mean Square	F Value	Pr > F
trt	8	51.75790299	6.46973787	2.55	0.0197
litter	7	69.96886926	9.99555275	3.94	0.0016

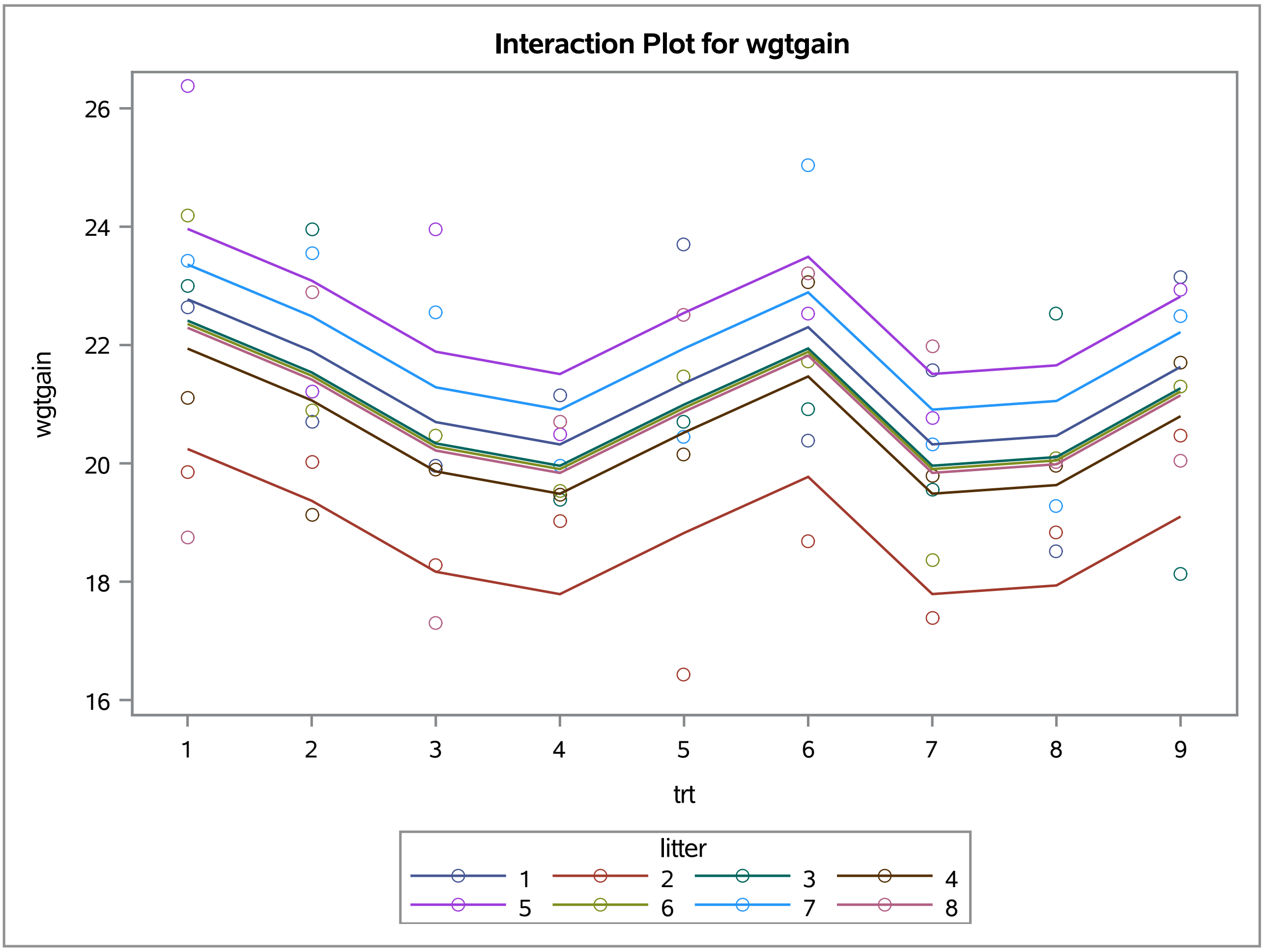


Table 7. Comparison: Traditional ANOVA (Litter as Fixed Effect)

Source	DF	Sum of Squares	Mean Square	F Value	p Value
trt	8	54.98293784	6.87286723	2.712	0.0140
litter	7	69.96886926	9.99555275	3.944	0.0016
trt	8	51.75790299	6.46973787	2.553	0.0197
litter	7	69.96886926	9.99555275	3.944	0.0016